Evidence-Adaptive Retrieval for OCR-Resilient Document Question Answering

REVISED EDITION — Research-Grade Re-Validation and Correction (2026-10-21)

Manthira Moorthy P · Vijay S K

IV ECE-A

Mentor: Dr. V. Karthikeyan, AP(Sl.G.)

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This edition supersedes the original QuIM_RAG_Research_Paper.docx. It was produced by a dedicated re-validation pass that treated no prior claim as trusted: it re-extracted the original paper's protocol, re-selected and corrected the benchmark dataset, froze a new experimental protocol before inspecting results, and re-ran the project's central claim under corrected conditions. That re-run found the original headline claim does not hold up once a known data-leakage issue is fixed (Section 11). This is reported here directly, per the explicit governing rule of this pass: evidence first, claims second; a rigorous negative or mixed result is preferable to an impressive but invalid comparison. Full audit trail: research/FINAL_VALIDATION/.

Abstract

Retrieval-augmented QA systems, including the QuIM-RAG baseline this project extends (Saha, Saha & Malik, IEEE Access, 2024), retrieve top-k passages and generate an answer unconditionally regardless of retrieval confidence. We extend an existing OCR-aware hybrid-retrieval prototype with real BM25 lexical scoring, real per-chunk OCR confidence captured from Tesseract, a residual vector quantizer (RVQ), and Evidence-Adaptive Retrieval & Verification (EAR): a logistic combination of four zero-extra-inference signals — dense/BM25 retrieval agreement, OCR confidence, cross-passage numeric contradiction, and query complexity — calibrated on held-out data to decide whether to answer, retrieve more, inspect structure, or abstain. A dedicated re-validation pass (this edition) found that the calibration/evaluation split used for every prior measurement of EAR's core claim was document-overlapping — 51% of calibration documents also appeared in evaluation — which inflates the apparent effect and understates its true uncertainty. Under a corrected, genuinely document-disjoint split on the OHR-Bench benchmark (Zhang et al., ICCV 2025), restricting answers to EAR's confident decisions still trends positive (62.3% → 68.4% top-1 accuracy, +6.1 points) but the coverage of confident decisions collapses from 62.8% to 15.4%, and the 95% bootstrap CI on the improvement widens to [−8.3, +19.7] points — it now includes zero, so the original claim of a statistically significant improvement (previously reported as 71.2% → 78.3%, CI [+2.5, +12.0]) is not supported under corrected conditions. We report this as a negative/inconclusive result for EAR's core claim, per this project's own evidence-first governing rule, rather than restating the original number. Separately, RVQ's recall gap remains root-caused to quantization reconstruction error (Welch's $t = -4.71$, $p = 3.5\times10^{-6}$) and 79% recovered by a quantize-then-exact-rerank fix — this analysis has no calibration/evaluation split and is unaffected by the leakage finding. The OCR-confidence/CER calibration result (Pearson $r = -0.98$) is similarly unaffected. The RAGAS generation-quality comparison (Faithfulness 0.514 → 0.619) and the NDSU-substitute reproduction (84.3% → 96.0%) both depend on the same uncorrected split family as the core claim and are flagged here as unverified under correction, not confirmed, pending a follow-up re-run. EAR's signal-computation overhead (≈95 ms, reusing retrieval already performed) remains roughly two orders of magnitude cheaper than an equivalent LLM-judge verification step (10.6 s measured mean) regardless of the above, since that comparison is architectural, not sample-dependent.

Contents

1. Introduction and Motivation
2. Related Work
3. Research Gap
4. What This Revision Changes — Summary
5. Problem Formulation
6. Proposed Method: Evidence-Adaptive Retrieval & Verification
7. System Architecture
8. Standard Algorithms: Formal Definitions
9. Experimental Setup (Corrected Protocol)
10. Datasets (Corrected)
11. Results: Corrected Main Benchmark — the Central Finding
12. Ablation Studies (Prior Split, Caveated)
13. Study: OCR Robustness and EAR Validation (Not Re-Run This Pass)
14. Study: End-to-End Generation Quality (Unverified Under Correction)
15. Study: RVQ Recall Gap Analysis (Unaffected by Correction)
16. Study: NDSU Substitute-Data Reproduction (Unverified Under Correction)
17. Statistical Rigor of This Revision
18. Limitations
19. Threats to Validity
20. Reproducibility Statement
21. Conclusion and Future Work
References

1. Introduction and Motivation

Retrieval-Augmented Generation (RAG) systems answer questions by retrieving relevant passages from a document corpus and conditioning a language model's output on them, rather than relying solely on the model's parametric memory. This improves factual grounding in principle, but in practice every RAG pipeline we examined — including the baseline this project extends — retrieves some passages and then unconditionally generates an answer, regardless of whether the retrieval was actually correct, whether OCR extraction was degraded, or whether the retrieved passages contradict each other.

Real-world document QA over scanned, OCR'd, and structurally complex documents is exactly where this failure mode is most likely. This project's central hypothesis is that a small set of signals, computable from retrieval that has already happened, can predict whether retrieved evidence is trustworthy well enough to justify withholding or qualifying an answer when it is not — without the latency and cost of an additional LLM call to make that judgment. This revision tests that hypothesis more rigorously than the original manuscript did, and reports what survives rigor and what does not.

2. Related Work

The direct baseline is QuIM-RAG (Saha, Saha & Malik, 2024), which improves retrieval by matching queries against LLM-generated candidate questions attached to each chunk, evaluated on a private NDSU corpus. Neither the original paper nor its baseline includes any mechanism for withholding an answer when evidence is insufficient, any lexical (BM25) retrieval path, or any treatment of OCR noise. A broader 2026 literature independently establishes confidence-gated abstention in RAG as an active research area (SURE-RAG-style verification models, evidence-sufficiency abstention benchmarks, retrieval-disagreement signals). This project's contribution, EAR, is situated within this area.

3. Research Gap

An audit of this project's own prior implementation identified two unimplemented items against its own roadmap — real BM25 and RVQ — both closed. Separately, neither the baseline paper nor the prior implementation included any decision layer for withholding an answer when retrieval confidence is low; this is the gap EAR addresses. A second, distinct research-integrity gap — found only during this revision's re-validation pass, not during the original research — is the subject of Section 4.

4. What This Revision Changes — Summary

A dedicated re-validation pass (full record: research/FINAL_VALIDATION/) treated no number in the original manuscript as trusted and re-checked each one against its source experiment file. It found:

- The benchmark dataset file used throughout (qas.json, 4,598 records) is not the file OHR-Bench's own documentation names as current (qas_v2.json, 8,498 records). Corrected in this revision.
- The calibration/evaluation split EAR's weights were fit and reported on was disjoint at the query-row level but NOT at the document level — 52 of 102 calibration documents (51%) also appeared

in the evaluation set. A document-level-disjoint split was built and verified (zero document and zero gold-page overlap) and used for a fresh re-run of the project's central claim.
- Under the corrected split, EAR's reported statistically-significant improvement (71.2%→78.3%, CI excludes zero) does not replicate: coverage collapses to 15.4% and the CI on the improvement widens to include zero (Section 11). This is reported as the paper's new central, honest finding, not hidden or minimized.
- A near-zero fitted weight on the ocr_confidence signal — present but previously under-examined — is now explicitly documented as making that signal functionally inert inside the combined model, despite being individually well-calibrated (Section 12).
- RVQ's and the OCR-degradation study's findings are unaffected by the leakage issue (neither uses that split) and are retained as previously reported, with that independence now stated explicitly rather than assumed.
- The RAGAS generation-quality and NDSU-reproduction studies share the same uncorrected split family as the core claim and are explicitly flagged as unverified under correction — not re-run this pass, not re-affirmed.

> **Per this revision's own governing rule: nothing below has been reworded to sound better than the data supports. Where a result weakened or reversed under correction, it is reported as weakened or reversed.**

5. Problem Formulation

Given a query q and a corpus indexed for hybrid (dense + BM25) retrieval, standard RAG always generates an answer from the top-k retrieved evidence set E_k(q). This project asks whether a decision function

$$f(q, E_k(q)) \longrightarrow \{answer, expand, inspect, abstain\} \tag{1}$$

computed from signals already available at retrieval time — with no additional embedding or language-model inference — can improve the expected correctness of answered queries over the unfiltered baseline, measured as a selective-prediction (risk–coverage) problem rather than as a retrieval-ranking metric.

6. Proposed Method: Evidence-Adaptive Retrieval & Verification (EAR)

EAR is a decision layer evaluated after Hybrid retrieval has already produced a candidate evidence set, not a replacement ranking function. It computes four signals, each normalized to [0,1], then combines them via a calibrated logistic model into a sufficiency score and a four-way decision. The method itself is unchanged by this revision — what changed is how rigorously its claimed benefit was measured (Section 11).

6.1 Signal 1 — Retrieval Agreement

$$a = |D_K \cap L_K| \,/\, |D_K \cup L_K| \tag{2}$$

6.2 Signal 2 — OCR Confidence

$$o = (1/K) \cdot \Sigma_\{c \in E_K\} \; conf(c)$$

(3)

6.3 Signal 3 — Cross-Passage Contradiction

$$x = |\{(i,j) : overlap(i,j) > 0.3 \wedge nums(i) \cap nums(j) = \varnothing\}| \; / \; C(K,2)$$

(4)

6.4 Signal 4 — Query Complexity

$$q = mean(\; min(tokens,30)/30,\; min(wh,3)/3,\; min(conj,3)/3 \;)$$

(5)

6.5 Sufficiency Score and Decision

$$z = w_0 + w_1 a + w_2 o + w_3 x + w_4 q$$

(6)

$$p = \sigma(z) = 1 \; / \; (1 + e\text{^}(-z))$$

(7)

θ_abstain and θ_answer are fit on the calibration split, never the evaluation split. This separation was enforced in both the original and the corrected implementation — the problem this revision found is in how the two splits were constructed (document overlap), not in whether fit and evaluation data were kept apart.

6.6 Calibrated weights — original split vs. corrected split

Table 1. *Logistic regression coefficients. Agreement and complexity remain the dominant weights in both fits; contradiction's weight shrinks substantially under the corrected split; OCR confidence remains uninformative in both.*

Signal	Weight, original (leaky) split	Weight, corrected (doc-disjoint) split
Agreement (a)	+1.664	+1.696
OCR confidence (o)	−0.003 (≈0)	−0.0003 (≈0, even smaller)
Contradiction (x)	−0.579	−0.033 (much weaker)
Complexity (q)	+1.030	+1.292

θ_abstain/θ_answer: 0.628/0.678 (original) vs. 0.503/0.752 (corrected) — the corrected thresholds are considerably further apart, which is the direct mechanism behind the coverage collapse reported in Section 11 (fewer evaluation queries fall above the now-higher θ_answer).

6.7 Why EAR remains computationally efficient regardless of Section 11's finding

This argument is architectural, not sample-dependent, so it is unaffected by the correction: EAR's only non-trivial cost is two extra retrieve() calls for the agreement signal; all other signals are O(K) or $O(K^2)$ arithmetic on data already in memory. Measured: ≈95 ms total EAR overhead vs. 10.6–15.1 s

for an LLM-judge verification alternative — the per-query generation/judging latencies themselves are unaffected by which calibration split EAR's decision weights were fit on.

Table 2. *EAR's measured efficiency vs. an LLM-judge alternative — unaffected by the Section 11 correction.*

Approach	Latency per decision	Extra model calls
EAR (this work)	≈95 ms	0 (reuses existing retrieval)
LLM-judge verification (this project's own RAGAS implementation)	10.6 – 15.1 s	5–6 per query

7. System Architecture

PDF / scanned document
→ PyMuPDF text extraction, else Tesseract OCR (+ real per-word confidence capture)
→ paragraph-aware chunking (950 characters, 100-character overlap)
→ MiniLM embeddings (384-dim; TF-IDF fallback) + real BM25 index, built side by side
→ retrieval: Dense cosine | BM25 | Hybrid (Reciprocal Rank Fusion, k=60) [+ optional cross-encoder rerank]
→ EAR: compute 4 signals → sufficiency score → decision
- answer → generate (local LLM / Anthropic API / extractive) + groundedness pre-check
- expand_retrieval → re-retrieve with larger k, then answer
- inspect_structure → surface conflicting evidence for human review
- abstain → no answer generated

Unchanged by this revision — only the data layer feeding EAR's calibration was corrected.

8. Standard Algorithms: Formal Definitions

Unchanged from the original manuscript; reproduced here for completeness. See research/paper/draft.md for the full derivations of BM25, dense cosine retrieval, Reciprocal Rank Fusion, cross-encoder reranking, Residual Vector Quantization, and the bootstrap/Welch/Pearson statistical tests used throughout.

9. Experimental Setup (Corrected Protocol)

research/run_benchmark_v2.py, implementing the frozen protocol in research/FINAL_VALIDATION/EXPERIMENT_PROTOCOL.md, written and committed to before this revision's new result was inspected. Fixed hyperparameters (k=5, 950-char/100-overlap chunking, all-MiniLM-L6-v2 embeddings, BM25 defaults) are unchanged from the original study — only the dataset file and the split construction changed.

10. Datasets (Corrected)

qas_v2.json (8,498 records — the file OHR-Bench's own README names as current), not qas.json (4,598 records, used throughout the original manuscript). Domains: law, finance, textbook, manual, academic (substituted for "paper", which has zero records in qas_v2.json's taxonomy), news. A stratified sample (≈70/domain target, seed 42) is partitioned by document, not by query row, into calibration (173 queries, 109 documents) and evaluation (247 queries, 153 documents) — verified

zero document overlap and zero gold-(document,page) overlap between the two pools, the pre-registered success criterion for this fix.

> The NDSU-substitute reproduction sample (70 queries, seed 17) and the RAGAS generation-quality sample (60 queries, seed 23) still use the original qas.json-derived data and the original, uncorrected split construction. They are retained below with that caveat attached, not silently upgraded.

11. Results: Corrected Main Benchmark — the Central Finding

247 evaluation queries, document-disjoint from the 173-query calibration split EAR was fit on. k=5.

Table 3. *Baseline retrieval comparison, corrected split. Hybrid-RRF's advantage over Dense-only replicates under correction; BM25 vs. Dense does not reach significance here (it did in the original, leaky-split run — see Table 4 note below).*

Config	Recall@5	nDCG@5	MRR@5	Latency (mean)	p vs. Dense (Bonferroni α=0.025)
Dense only	0.427	0.478	0.683	36.4 ms	—
BM25 (real)	0.426	0.492	0.710	36.9 ms	0.383 (n.s.)
Hybrid (RRF)	0.444	0.505	0.726	39.6 ms	0.008 (significant)

> **Reversal vs. the original manuscript: the original (leaky-split) run reported BM25 alone (nDCG@5=0.528) beating Hybrid RRF (0.515) as a notable finding. Under the corrected, larger, document-disjoint evaluation set, this reverses — Hybrid (0.505) now exceeds BM25 (0.492), and Hybrid is the config that reaches significance vs. Dense while BM25 does not. The original 'BM25 beats Hybrid' claim should not be repeated.**

11.1 EAR selective-prediction result — original claim vs. corrected re-run

Table 4. *The project's central claim, before and after correcting the calibration/evaluation split. Full derivation: research/FINAL_VALIDATION/FINAL_RESULTS.md Section 2.*

Quantity	Original (leaky split)	Corrected (document-disjoint)
Calibration / evaluation size	168 / 252 (51% doc. overlap)	173 / 247 (0% doc. overlap)
Coverage	62.8%	15.4%
Accuracy on answered	78.3%	68.4%
Unfiltered baseline accuracy	71.2%	62.3%
Improvement	+7.1 points	+6.1 points
95% bootstrap CI on improvement	[+2.5, +12.0] — excludes zero	[−8.3, +19.7] — includes zero
Statistically significant at α=0.05?	Yes, as originally reported	No

> **This is the headline finding of this revision. The original claim that EAR produces a statistically significant selective-accuracy improvement is NOT SUPPORTED once the known document-level leakage between calibration and evaluation is corrected. The point estimate of the improvement is numerically similar (+6.1 vs. +7.1 points), but usable coverage collapses by a factor of 4 (62.8%→15.4%) and the confidence interval on the improvement now includes zero. We report this as a negative/inconclusive result for the corrected condition, consistent with this project's own evidence-first governing rule, rather than restating the original number as if it still held.**

Mechanism: the corrected calibration fit produces a wider gap between θ_abstain (0.503) and θ_answer (0.752) than the original fit (0.628/0.678 — Table 1), so far fewer evaluation queries clear

the higher θ_answer bar, concentrating the 'answer' decision into a small (n=38), more favorably-selected subset whose accuracy estimate carries much wider sampling uncertainty.

12. Ablation Studies (Prior Split, Caveated)

A leave-one-signal-out ablation was not re-run on the corrected split this pass (scope boundary — see research/FINAL_VALIDATION/EXPERIMENT_PROTOCOL.md §9). The table below is the original (leaky-split) ablation, retained for its qualitative finding about the ocr_confidence signal, which independently corroborates in both the original and corrected weight fits (Table 1, Section 6.6).

Table 5. *Original (leaky-split) leave-one-signal-out ablation. Not re-validated on the corrected split.*

Ablation	Coverage	Accuracy on answered	Baseline
Full model (4 signals)	62.8%	78.3%	71.2%
without agreement	19.2%	75.0%	71.2%
without ocr_confidence	62.8% (identical)	78.3% (identical)	71.2%
without contradiction	63.6%	78.6%	71.2%
without complexity	40.4%	79.2%	71.2%

> Removing ocr_confidence changes nothing — identical coverage and accuracy to 13 decimal places. This confirms (independently of the leakage issue) that ocr_confidence's fitted weight is near-zero in both splits and is functionally inert inside the combined model, despite being individually well-calibrated against real character error rate (Section 13). A corrected ablation would very plausibly show at least one signal-removal configuration collapsing to near-zero coverage, given the full corrected model's own 15.4% coverage — flagged as a follow-up requiring its own minimum-coverage reporting floor, not run this pass.

13. Study: OCR Robustness and EAR Validation (Not Re-Run This Pass)

Unaffected by the Section 11 correction — this study uses no calibration/evaluation split at all; it compares real Tesseract confidence against real character error rate under synthetic degradation of real OHR-Bench ground truth text. Retained as previously reported.

Table 6. *OCR confidence vs. Character Error Rate, 45 degraded observations across 15 documents. Not affected by this revision's correction.*

Correlation	Coefficient	p-value
Pearson r (confidence vs. CER)	−0.984	1.4×10^{-30}
Spearman ρ	−0.911	1.4×10^{-16}

14. Study: End-to-End Generation Quality (Unverified Under Correction)

> **This study's EAR-gating decisions were computed using the original, uncorrected calibration/evaluation split family (same adapter as the original main benchmark). Given Section 11's finding, the coverage and accuracy figures below should be treated as UNVERIFIED under correction, not confirmed, until re-run against the document-disjoint split. Retained here, not deleted, so the gap is visible rather than hidden.**

Table 7. *RAGAS-style metrics, 60-query sample, local LLM generator and judge — unverified under correction (see caveat above).*

Metric	Baseline	Proposed (EAR-gated)	Δ
Faithfulness	0.514	0.619	+10.5 pts
Answer Relevancy	0.700	0.704	+0.4 pts
Context Precision	0.721	0.767	+4.6 pts
Context Recall	0.410	0.435	+2.5 pts

15. Study: RVQ Recall Gap Analysis (Unaffected by Correction)

A single eval-engine comparison with no calibration/evaluation split — architecturally immune to the Section 11 leakage finding. Retained as previously reported and re-confirmed by direct source-file inspection this session (research/FINAL_VALIDATION/CLAIM_VERIFICATION_MATRIX.md, claim #5).

Table 8. *Welch's t-test: $t = -4.71$, $p = 3.5 \times 10^{-6}$. Confirmed by direct re-inspection of the source JSON this revision session.*

Group	n	Mean reconstruction cosine
Lost to quantization	276	0.621
Not lost	1,219	0.671
Corpus-wide average	16,076	0.761

Table 9. *RVQ quantize-then-exact-rerank recovers 79% of lost Recall@5 at statistically the same latency.*

Config	Recall@5	MRR@5	Latency (mean)
Exact dense (no quantization)	0.378	0.726	48.3 ms
RVQ quantized (current, no rerank)	0.161	0.276	12.9 ms
RVQ quantized + exact rerank (fix)	0.301	0.611	13.0 ms

16. Study: NDSU Substitute-Data Reproduction (Unverified Under Correction)

> **This study's EAR-gating result (84.3%→96.0%) was offered in the original manuscript as a second, independent confirmation of the main benchmark's EAR claim. Given Section 11's finding that the main claim's own calibration/evaluation split was leaky, and that this reproduction's split was built with the same un-corrected adapter, this second-confirmation claim is also UNVERIFIED under correction. It is not, by itself, proof of leakage in this specific 70-query sample (which was never checked for document overlap either way this session), but it can no longer be cited as independent corroboration of Section 11's original (now-superseded) number.**

Table 10. *Substitute-data retrieval comparison, 70 queries. The retrieval-only comparison (not EAR-dependent) is unaffected by the correction and is retained as previously reported.*

Config	Recall@5	MRR@5	nDCG@5
Traditional RAG (paper's baseline)	0.710	0.815	0.713

QuIM-style (reproduction)	0.344	0.459	0.341
This project: Hybrid RRF	0.705	0.879	0.732

The retrieval-only finding — that this environment's question-matching reproduction underperforms plain dense retrieval, plausibly due to a weaker question generator than the original paper's GPT-3.5 — is independent of EAR and is unaffected by this revision's correction.

17. Statistical Rigor of This Revision

Full detail: research/FINAL_VALIDATION/STATISTICAL_ANALYSIS.md. Summary: Bonferroni correction (α=0.025) applied to the 2 baseline-retrieval pairwise tests in Section 11; Hybrid-RRF's advantage over Dense survives correction (p=0.008), BM25's does not (p=0.383). The single EAR-vs-baseline comparison is the pre-registered primary endpoint and is not folded into that correction family. The corrected EAR CI is markedly wider than the original primarily because the 'answer' bucket shrank from n=157 to n=38 — a direct, measurable consequence of the corrected, stricter calibration fit, not an artifact of the bootstrap procedure itself. EAR signal-level ROC-AUC/Brier/calibration-reliability diagnostics, requested by the governing validation mandate, were not implemented this pass and remain the single highest-value follow-up.

18. Limitations

- EAR's core selective-accuracy claim does not reach statistical significance under a corrected, document-disjoint calibration/evaluation split (Section 11) — this is now this project's primary limitation, not a secondary caveat.
- The RAGAS generation-quality study and the NDSU-substitute reproduction share the same uncorrected split family as the original main-benchmark claim and have not themselves been re-verified under correction.
- EAR's ocr_confidence weight remains uninformative in every calibration run performed to date (original and corrected) — a calibration-design issue (narrow OCR-confidence variance in the sampled data), not necessarily a signal-quality issue, but not yet fixed and tested.
- RAGAS metrics are this project's own implementation, judged by a 3.2B-parameter local model, not validated against a cloud-scale judge.
- Single primary benchmark family (OHR-Bench and its sampled subsets), single embedding model (MiniLM), single OCR engine (Tesseract) throughout — both before and after this revision.
- No repeated-seed variance estimate across different random splits anywhere in this project — all reported CIs capture only within-split bootstrap variance, not split-to-split variance.

19. Threats to Validity

The corrected calibration and evaluation splits are disjoint by document construction but are still drawn from the same underlying OHR-Bench sample family, not independently collected corpora. The NDSU-substitute and RAGAS samples were not checked this session for the same document-overlap property found in the main benchmark's original split — their own leakage status is simply unknown, not confirmed clean.

20. Reproducibility Statement

This revision's one new experiment is reproducible with:

```
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 .venv/bin/python research/run_benchmark_v2.py
```

Seed 42 throughout. Full environment, exact commands, and the HuggingFace-offline-mode fix required to avoid a network hang are documented in research/FINAL_VALIDATION/REPRODUCIBILITY_GUIDE.md, including a pip freeze snapshot. Test suite: 70 passed, 0 failed (67 pre-existing + 3 new regression tests added this session for the document-disjoint-split fix, in tests/test_ohrbench_adapter_v2.py).

21. Conclusion and Future Work

This revision's primary contribution is methodological: it found and corrected a document-level data-leakage bug that had inflated the project's central novel-contribution claim past the point of statistical significance, and it reports the corrected, weaker result directly rather than preserving the stronger original number. EAR's effect, under correction, trends positive but is not statistically distinguishable from no effect at the pre-registered significance level, at substantially lower coverage than previously reported. RVQ's root-cause analysis and partial fix, and the OCR-confidence calibration result, are unaffected by this finding and stand as this project's most solidly supported contributions. Priority next steps, in order of expected value: (1) check the NDSU-substitute and RAGAS samples for the same document-overlap issue and re-run both if found; (2) implement EAR signal-level ROC-AUC/Brier/reliability diagnostics to determine which, if any, individual signal retains predictive value under the corrected split; (3) re-calibrate EAR with a larger or differently-constructed calibration set, since the corrected fit's 15.4% coverage is likely too restrictive to be practically useful even if statistically borderline; (4) extend the document-disjoint-split regression test added this session to cover the NDSU and RAGAS sample builders too.

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Full audit trail for this revision: research/FINAL_VALIDATION/ (BASELINE_AUDIT.md, ORIGINAL_PAPER_PROTOCOL.md, DATASET_SELECTION.md, EXPERIMENT_PROTOCOL.md, FINAL_RESULTS.md, STATISTICAL_ANALYSIS.md, ABLATION_ANALYSIS.md, FAILURE_ANALYSIS.md, REPRODUCIBILITY_GUIDE.md, CHANGES_AND_FIXES.md, CLAIM_VERIFICATION_MATRIX.md).